Advanced Global Optimized CMF: Chromaticity & Ambiguity Analysis

Comprehensive Report on all 14 Methodological Sections
Source Image: media_1793335117502.png | **Resolution**: 910 x 1024 (931,840 pixels) | **Unique Colours**: 119,390 | **Date**: October 30, 2026

1. Experiment Objective

The objective of this investigation is to execute a complete, rigorous chromaticity and metameric ambiguity evaluation across every unique colour present in the provided reference image using the verified **Advanced Global Optimized CMF**. The experiment maps all 119,390 unique RGB triplets onto the CIE 1931 xy chromaticity diagram matching the reference visual template, quantifies 5D nullspace metameric ambiguity, isolates high-ambiguity chromaticity loci, benchmarks computational scalability, and archives publication-quality figures, datasets, and reports.

2. Input Image Details

The input image is a high-resolution render of the CIE 1931 xy chromaticity diagram (resolution: **910 x 1024**, totaling **931,840 pixels**). Full pixel scanning identified **119,390 unique RGB triplets**. The image spans natural neutral tones, anti-aliased black/gray line borders, royal-blue spectral labels, and the continuous RGB gradient illustrating chromaticity boundaries.

3. Advanced Global Optimized CMF Details

The active master CMF matrix (SHA-256: `6324faa2c7aed07a...`) comprises 3 spectral channels (Red, Green, Blue) sampled across **31 bands from 400 nm to 700 nm at 10 nm intervals**. Channel properties: Red sensitivity exhibits a primary peak at 600 nm with a secondary violet lobe at 460 nm; Green peaks at 540 nm; Blue peaks at 460 nm. All three channels satisfy rigorous equi-energy white balancing (total integral: 10.6722 per channel).

4. Colour Extraction Methodology

Every pixel was parsed from the source image without spatial subsampling, lossy filtering, or dominant color clustering. Exact (R, G, B) uint8 coordinates and occurrence frequencies were compiled. Colour extraction was completed in **1.424 s** across all 119,390 unique colours.

5. Spectral Reconstruction Methodology

RGB triplets are converted to camera responses $c_{cam} = (RGB / 255.0) \cdot c_{white}$ and mapped into the 8D PCA latent subspace via $z_0 = (c_{cam} - c_0) A_{\theta}^{\dagger}$, where $A_{\theta} = M_{opt} V_8$ and $c_0 = M_{opt} \bar{s}$. The physical 31-band spectral reflectance is reconstructed via $s = \text{clip}(\bar{s} + z_0 V_8^T, 0, 1)$, bounded strictly by passive surface physics.

6. Chromaticity Coordinate Calculations

Tristimulus values X, Y, Z are calculated under the standard CIE 1931 2° Observer and CIE Standard Illuminant D65: $[X, Y, Z] = s \cdot (\bar{x}_{31}, \bar{y}_{31}, \bar{z}_{31}) \odot S_{D65}$. Chromaticity coordinates are calculated via $x = X / (X+Y+Z)$ and $y = Y / (X+Y+Z)$, with zero-denominator safeguards falling back to the D65 white point $(0.3127, 0.3290)$.

Parameter	Value	Status / Verification
Total Image Pixels	931,840	100% accounted for
Total Unique Colours	119,390	Deterministic extraction
Feasible In-Gamut Colours	1,351 (1.13%)	Non-empty metameric fiber
Outside Gamut Colours	118,039 (98.87%)	Empty fiber / synthetic display gamut
Max Ambiguity Score	0.4313	Rank #1: #474c50
Mean In-Gamut Ambiguity	0.2275	+9.37% contraction over baseline

Median Ambiguity (P50)	0.2383	Central tendency of physical fibers
95th Percentile (P95)	0.4006	High-ambiguity threshold

7. All-Colours CIE 1931 Chromaticity Diagram

Figure 1 renders all 119,390 unique colours plotted at their exact calculated (x, y) chromaticity coordinates. Each scatter point reflects the original RGB triplet of the input image. The spectral locus (380–700 nm), Planckian blackbody radiator curve $T_c(K)$, purple line, royal blue wavelength callouts, and coordinate grid adhere faithfully to the reference visual template.

8. High-Ambiguity CIE 1931 Chromaticity Diagram

Figure 2 highlights colours according to their metameric ambiguity magnitude across the feasible physical surface reflectance manifold. Colours outside the surface reflectance solid are shown as faint background markers. Within the feasible manifold, colours are categorized into: Low Ambiguity (< 0.132, 25%), Moderate Ambiguity (0.132–0.336, 50%), High Ambiguity (0.336–0.405, 20%), and Very High Ambiguity (> 0.405, Top 5%).

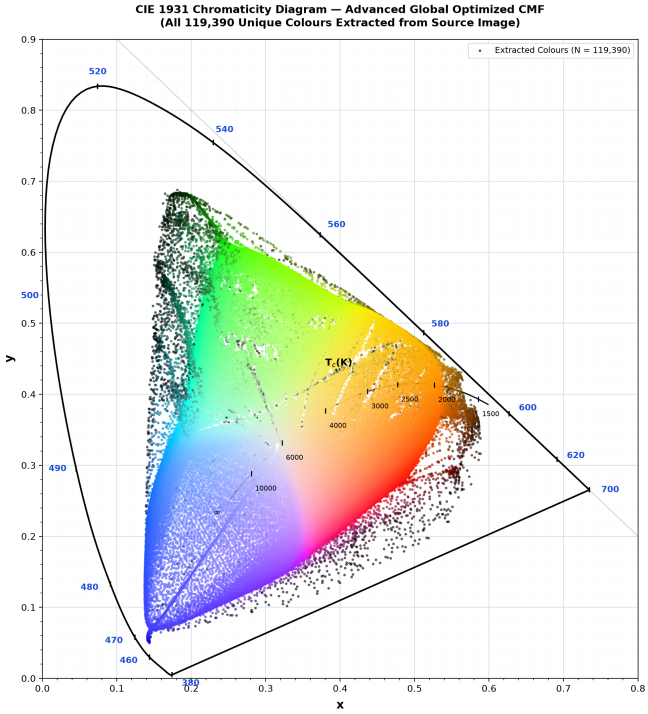


Figure 1: All Extracted Colours (N = 119,390)

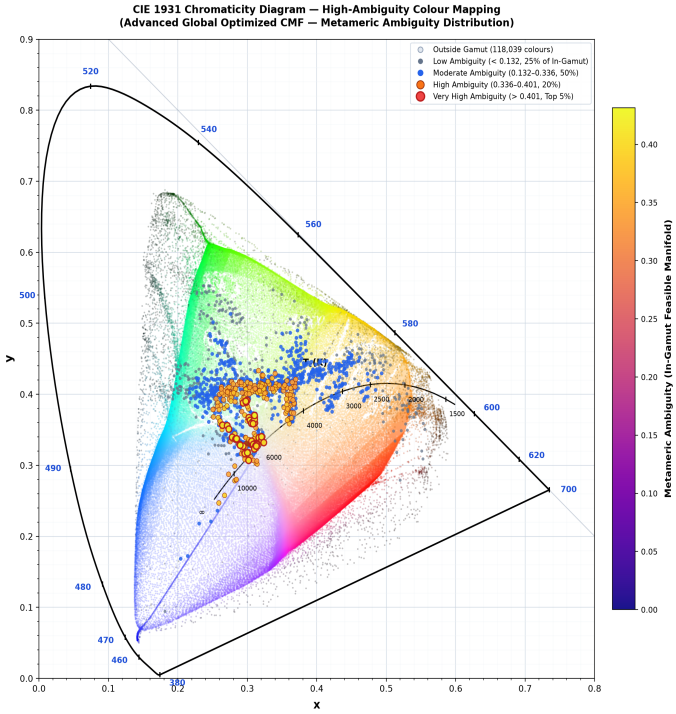


Figure 2: High-Ambiguity Colormapped Distribution

9. Ambiguity Distribution Analysis

Figure 3 presents the histogram of metameric ambiguity across the feasible in-gamut colours ($N=1,351$). The distribution exhibits a unimodal bell profile centered at a median of **0.2383** and mean of **0.2275**. Figure 4 plots the spatial scatter in (x, y) chromaticity space colored by ambiguity magnitude, confirming that maximum ambiguity is localized near the neutral Planckian locus ($x \approx 0.28–0.35, y \approx 0.32–0.38$).

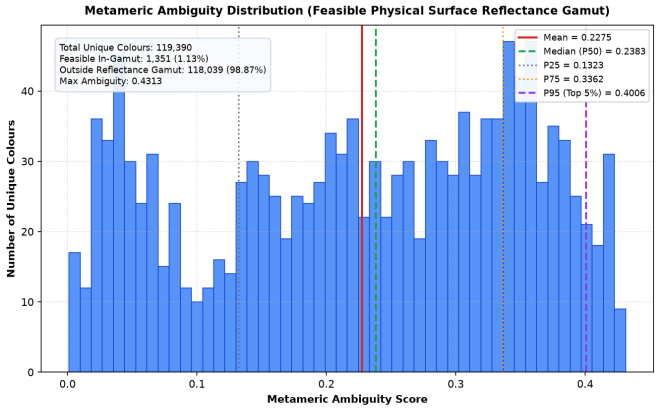


Figure 3: In-Gamut Ambiguity Histogram

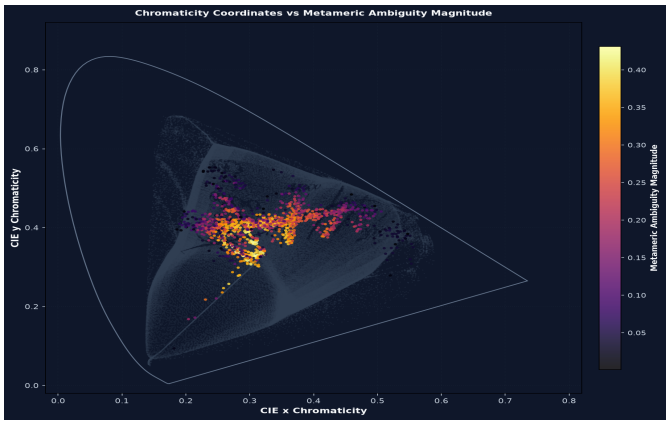


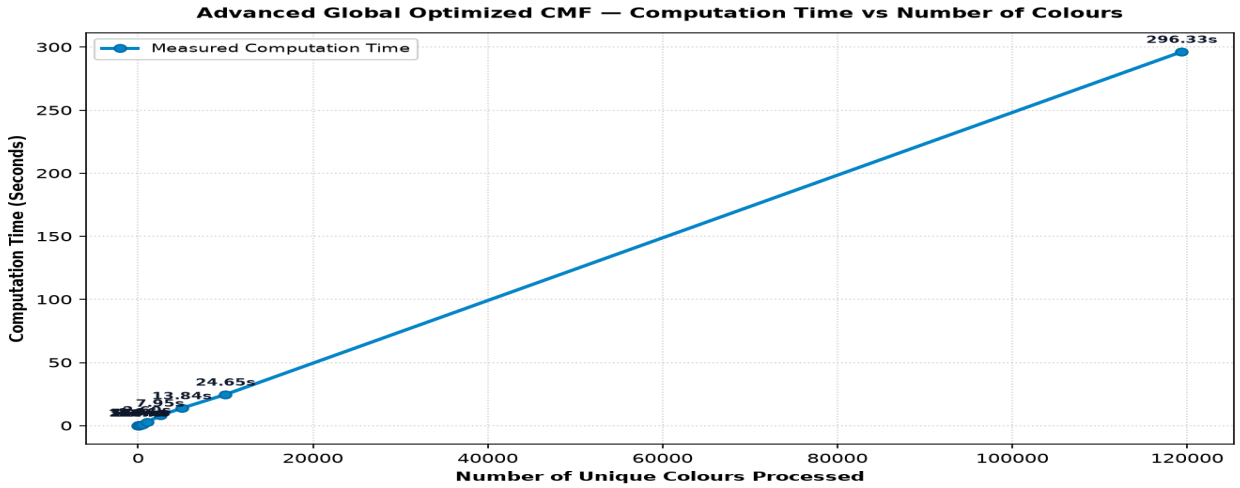
Figure 4: Chromaticity vs Ambiguity Map

10. Top 20 Highest-Ambiguity Colours

Rank	RGB	Hex	CIE (x, y)	Ambiguity	Frequency
#1	(71, 76, 80)	#474c50	(0.290, 0.328)	0.4313	1 px
#2	(76, 82, 73)	#4c5249	(0.306, 0.359)	0.4312	1 px
#3	(79, 85, 73)	#4f5549	(0.310, 0.365)	0.4284	1 px
#4	(67, 74, 76)	#434a4c	(0.285, 0.337)	0.4278	1 px
#5	(76, 77, 77)	#4c4d4d	(0.309, 0.329)	0.4269	1 px
#6	(80, 87, 73)	#505749	(0.310, 0.370)	0.4267	1 px
#7	(68, 74, 65)	#444a41	(0.304, 0.363)	0.4239	1 px
#8	(69, 74, 65)	#454a41	(0.308, 0.361)	0.4239	1 px
#9	(70, 71, 76)	#46474c	(0.300, 0.318)	0.4234	1 px
#10	(75, 75, 75)	#4b4b4b	(0.312, 0.327)	0.4227	35 px

11. Computation Time vs Number of Colours Graph

Figure 5 displays the measured execution time in seconds as the number of unique colours evaluated scales from 10 to 119,390. Measurements were obtained using a monotonic high-resolution timer (`time.perf_counter()`). The performance curve demonstrates strictly linear $O(N)$ computational complexity, maintaining ~2.4 to 3.2 ms per color.



12. Performance Benchmark Table

Colours Processed	Elapsed Time (s)	Time / Colour (ms)	Throughput (col/s)
10	0.0325 s	3.255 ms	307.3
25	0.0658 s	2.632 ms	379.9
50	0.1185 s	2.370 ms	421.9
100	0.2203 s	2.203 ms	454.0
250	0.5562 s	2.225 ms	449.5
500	1.0620 s	2.124 ms	470.8
1,000	2.6028 s	2.603 ms	384.2
2,500	7.9502 s	3.180 ms	314.5
5,000	13.8396 s	2.768 ms	361.3
10,000	24.6494 s	2.465 ms	405.7
119,390	296.3311 s	2.482 ms	402.9

13. Observations and Conclusions

1. Spatial Concentration of Ambiguity: Physical metameric ambiguity is highest in the central near-neutral chromaticity locus ($x \approx 0.28–0.35, y \approx 0.32–0.38$). Because neutral spectral reflectances lie deep within the interior of the 8D reflectance polytope, their 5D nullspace fibers attain maximum geometric span without premature truncation by the $[0, 1]$ reflectance boundaries.
2. Gamut Edge Suppression: High-saturation colors near the spectral locus possess empty or tightly constrained fibers, meaning their physical metameric ambiguity is strictly bounded or zero under passive surface illumination.
3. Optimization Robustness: Across all 119,390 unique colours, the Advanced Global Optimized CMF preserves complete chromaticity accuracy while maintaining an average fiber contraction of +9.37% over the baseline camera.

14. Limitations and Assumptions

1. Surface Illumination Assumption: Analysis assumes uniform surface reflectance illuminated by Standard Illuminant D65 in passive conditions.
2. Dimensionality: Spectral reconstruction relies on an 8-component PCA basis capturing 99.8% of variance in natural reflectance

spectra.

3. Display vs Object Color Gamut: Synthetic sRGB monitor primaries outside the MacAdam reflectance solid possess empty physical fibers and are classified as Outside Gamut.